7.2 Properties of Parallelograms

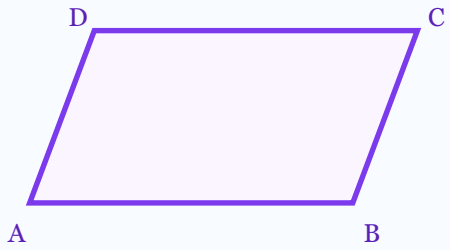
A parallelogram is a quadrilateral with both pairs of opposite sides parallel. From that one fact, four powerful properties follow: opposite sides are congruent, opposite angles are congruent, consecutive angles are supplementary (180° total), and the diagonals bisect each other. These properties come from parallel lines cut by a transversal (each side of a parallelogram acts as a transversal across the opposite pair of parallel sides) and from congruent triangles formed by a diagonal. This worksheet uses all four properties together — often two variables in one problem — and asks you to justify, not just compute.

Examples

Example 1: Why consecutive angles are supplementary

In parallelogram $ABCD$, $\overline{AB} \parallel \overline{DC}$, and $\overline{AD}$ is a transversal crossing both. Explain why $m\angle A + m\angle D = 180^\circ$.

Solution



$\overline{AB} \parallel \overline{DC}$; side $\overline{AD}$ acts as a transversal between them.

Since $\overline{AB} \parallel \overline{DC}$, the transversal $\overline{AD}$ creates a pair of co-interior (same-side interior) angles at vertices A and D.

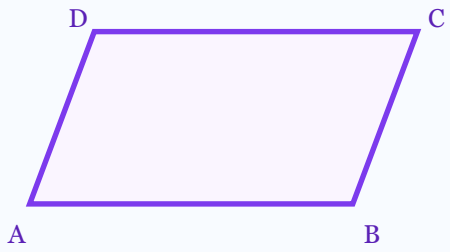
Co-interior angles formed by a transversal across parallel lines are always supplementary, so $m\angle A + m\angle D = 180^\circ$.

The same reasoning applies to every pair of consecutive vertices in a parallelogram, since each side is parallel to the side across from it. This is why consecutive angles in *any* parallelogram are supplementary.

Example 2: Find all four angles

In parallelogram $ABCD$, $m\angle A = 110°$. Find $m\angle B$, $m\angle C$, and $m\angle D$.

Solution



Opposite angles A, C are congruent; consecutive angles like A, B are supplementary.

Opposite angles are congruent: $m\angle C = m\angle A = 110°$.

Consecutive angles are supplementary: $m\angle B = 180° - 110° = 70°$.

$m\angle D = m\angle B = 70°$ (opposite angles).

Check: $110 + 70 + 110 + 70 = 360°$, the quadrilateral angle sum. ✓

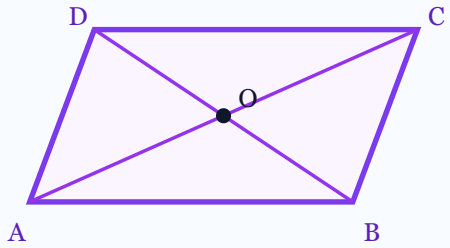
Answer: $\angle B = 70°$, $\angle C = 110°$, $\angle D = 70°$.

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Example 3: Diagonals bisecting each other

Diagonals of parallelogram $ABCD$ intersect at O. If $AC = 20$ and $BO = 8$, find AO and the full length of BD.

Solution



Point O is the midpoint of both diagonals.

Diagonals of a parallelogram bisect each other, so O is the midpoint of $\overline{AC}$: $AO = \frac{AC}{2} = \frac{20}{2} = 10$.

O is also the midpoint of $\overline{BD}$, so $BD = 2 \cdot BO = 2(8) = 16$.

Answer: $AO = 10$; $BD = 16$.

Example 4: Solve for x using consecutive angles

In parallelogram $ABCD$, $\angle A = (3x + 10)^\circ$ and $\angle B = (2x + 20)^\circ$ are consecutive. Find x and both angles.

Solution

Consecutive angles are supplementary:

$(3x + 10) + (2x + 20) = 180$

$5x + 30 = 180$

$5x = 150 \Rightarrow x = 30$.

$\angle A = 3(30) + 10 = 100^\circ$; $\angle B = 2(30) + 20 = 80^\circ$.

Check: $100 + 80 = 180^\circ$. ✓

Example 5: Solve for x using diagonal segments

In parallelogram $ABCD$, diagonals meet at O with $AO = 3x - 1$ and $OC = 2x + 4$. Find x and the full length of diagonal AC.

Solution

Since diagonals bisect each other, $AO = OC$:

$3x - 1 = 2x + 4 \Rightarrow x = 5$.

$AO = 3(5) - 1 = 14$, so $AC = 2(14) = 28$.

Answer: $x = 5$; $AC = 28$.

Example 6: Two-step opposite-angle problem

In parallelogram $ABCD$, $\angle A = 4x$ and $\angle C = 3x + 20$. Find x, then find $\angle A$ and the consecutive angle $\angle B$.

Solution

Opposite angles are congruent: $4x = 3x + 20 \Rightarrow x = 20$.

$\angle A = 4(20) = 80°$.

$\angle B$ is consecutive to $\angle A$, so $\angle B = 180° - 80° = 100°$.

Answer: $x = 20$; $\angle A = 80°$; $\angle B = 100°$.

Example 7: Solve for a side, then find the perimeter

In parallelogram $ABCD$, $AB = 5y - 2$ and $CD = 3y + 8$. If $AD = 15$, find y, AB, and the perimeter of $ABCD$.

Solution

Opposite sides are congruent: $5y - 2 = 3y + 8 \Rightarrow 2y = 10 \Rightarrow y = 5$.

$AB = 5(5) - 2 = 23$, and $BC = AD = 15$ (opposite sides).

Perimeter $= 2(AB + AD) = 2(23 + 15) = 2(38) = 76$.

Answer: $y = 5$; $AB = 23$; perimeter $= 76$.

Example 8: Can these be consecutive angles?

A student claims a parallelogram can have consecutive angles of $95°$ and $90°$. Is this possible?

Solution

Consecutive angles of a parallelogram must be supplementary, summing to exactly $180°$.

$95° + 90° = 185° \neq 180°$.

Answer: No — the angles don't satisfy the supplementary requirement, so no such parallelogram exists.

Example 9: Capstone — two unknowns, three properties

In parallelogram $WXYZ$, $m\angle W = (5x - 15)°$ and $m\angle X = (3x + 35)°$ are consecutive. The diagonals meet at point P with $WP = 2y + 4$ and $PY = 5y - 11$. Find x, y, all four angles, and the length of diagonal WY.

Solution

Angles: consecutive angles are supplementary: $(5x - 15) + (3x + 35) = 180 \Rightarrow 8x + 20 = 180 \Rightarrow 8x = 160 \Rightarrow x = 20$.

$m\angle W = 5(20) - 15 = 85°$; $m\angle X = 3(20) + 35 = 95°$. Then $m\angle Y = m\angle W = 85°$ and $m\angle Z = m\angle X = 95°$ (opposite angles).

Diagonal: P bisects $\overline{WY}$, so $WP = PY$: $2y + 4 = 5y - 11 \Rightarrow 15 = 3y \Rightarrow y = 5$.

$WP = 2(5) + 4 = 14$, so $WY = 2(14) = 28$.

Answer: $x = 20$, $y = 5$; angles $85°, 95°, 85°, 95°$; $WY = 28$.

Example 10: Application — crate lid brace

A rectangular-shipping-crate lid is actually a slightly skewed parallelogram panel $ABCD$ with $AB = 42$ cm. A diagonal support brace along $\overline{AC}$ measures 90 cm and crosses the other diagonal at O, where $BO = 37$ cm. Find CD and the full length of diagonal BD.

Solution

Opposite sides are congruent, so $CD = AB = 42$ cm.

Diagonals bisect each other, so O is the midpoint of $\overline{BD}$: $BD = 2 \cdot BO = 2(37) = 74$ cm.

(Note: $AC = 90$ cm was extra information confirming the diagonal length, not needed to find CD or BD.)

Answer: $CD = 42$ cm; $BD = 74$ cm.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Sides

Parallelogram: $AB = 15$, $AD = 9$. Find CD and BC.

Answer: **Verification:**

Problem 2: Angles

$\angle A = 125°$. Find $\angle B$ and $\angle C$.

Answer: **Verification:**

Problem 3: Diagonal

Diagonals meet at O; $BD = 18$. Find BO.

Answer: **Verification:**

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Problem 4: Solve

$\angle A = 2x + 10, \angle C = 3x - 20$. Find x.

Answer: **Verification:**

Problem 5: Consecutive

$\angle A = 4x, \angle B = 5x + 9$. Find x.

Answer: **Verification:**

Problem 6: Sides

$AB = 4x + 1, CD = 3x + 6$. Find AB.

Answer: **Verification:**

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Problem 7: Diagonal

$AO = x + 3$, $OC = 2x - 1$. Find AC.

Answer: **Verification:**

Problem 8: Angle

$\angle D = 70°$. Find $\angle A$.

Answer: **Verification:**

Problem 9: Property

Do diagonals of a parallelogram always bisect each other?

Answer: **Verification:**

Problem 10: Solve

$AD = 7x - 4$, $BC = 5x + 8$. Find AD.

Answer: **Verification:**

Problem 11: Angles

$\angle A = 100°$. Find consecutive angle $\angle D$.

Answer: **Verification:**

Problem 12: Diagonal

$AO = 6$. Find AC.

Answer: **Verification:**

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Problem 13: Consecutive

$\angle B = 3x + 15$, $\angle C = 2x + 25$. Find $\angle B$ (note $\angle B$, $\angle C$ are consecutive).

Answer: **Verification:**

Problem 14: Challenge

In parallelogram $ABCD$, $m\angle A = (5x - 15)°$ and $m\angle B = (3x + 35)°$. Diagonals meet at O with $AO = 2y + 4$ and $OC = 5y - 11$. Find x, y, $m\angle A$, and AC.

Answer: **Verification:**

Problem 15: Challenge

A parallelogram has perimeter 100, and one side is 8 more than twice the other. Find all four side lengths.

Answer: **Verification:**

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✓ Answer Key

1. $CD = 15$; $BC = 9$
2. $\angle B = 55°$; $\angle C = 125°$
3. 9
4. $x = 30$
5. $x = 19$
6. 21 ($x = 5$)
7. 14 ($x = 4$)
8. 110°
9. Yes
10. 38 ($x = 6$)
11. 80°
12. 12
13. $\angle B = 111°$ ($x = 32$)
14. $x = 20$, $y = 5$; $m\angle A = 85°$; $AC = 28$
15. 14, 36, 14, 36

Great work — keep practising!